

KAPIL RAVI RATHOD

Salt Lake City, UT 84101

+1 (582) 203-8438 ✉ kapilrathod1234@gmail.com 📁 Portfolio [in](#) [Linkedin](#) [on](#) [GitHub](#)

EDUCATION

The Pennsylvania State University, University Park

Aug 2024 – May 2026 (Expected)

GPA: 3.9 | M.Sc. in Computer Science and Engineering

State College, PA

National Institute of Technology, Warangal

Aug 2017 – May 2021

CGPA: 7.98 | B.Tech. in Computer Science and Engineering

Warangal, TS

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, Go, R, JavaScript, Bash, CUDA C++

Relational Databases: MySQL, PostgreSQL, Oracle

NoSQL and Vector Databases: MongoDB, Redis, Qdrant, FAISS, Milvus

Frameworks & Dev Tools: Spring Boot, Angular.js, Node.js, React.js, Django, Docker, Git, Hadoop, gRPC, Kafka, Spark

Testing Frameworks: PyTest, unittest, Locust, Behave, Selenium, JUnit

Cloud & Orchestration Tools: Google Cloud Platform [Link], Prefect, Camunda, OpenShift, Jenkins, Kubernetes

AI/ML Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, LangChain, AutoGen, LangGraph, DSPy, DeepEval

EXPERIENCE

Recursion Pharmaceuticals

May 2025 – Aug 2025

SWE Intern

Salt Lake City, UT

- Built scalable data pipelines using Prefect to automate workflows and support critical stakeholder dashboards in Looker.
- Developed Django and React components for the Invivomics web app; managed deployments and data migrations.

Deutsche Bank

Jul 2021 – Jul 2024

Associate Software Engineer (SDE-2)

Pune, MH

- Architected a RAG-powered multi-LLM chatbot unifying internal docs; used by **900+ users** and saving **\$3M/year**.
- Engineered an event-driven LLM-based service for regulatory change alerts, **saving \$1M/year** and **700+ hours**.

Graduate Analyst (SDE-1)

- Automated UC4 pre-deployment checks with Python/UNIX to parse config files and flag changes, reducing manual effort.
- Refined a bank-wide translation service, **boosting the BLEU score by 12%** across intranet pages.
- Improved a reporting app's infra, security and UI; created dashboards that **saved 30+ hours/week** on manual updates.
- Built a BERT-based automated resume matching tool, **saving 80% of manual effort across 100+ hiring managers**.

Semut.io

Apr 2020 – Jul 2020

Cloud Native Engineering Intern

Remote

- Contributed to a Go-based application manager enabling streamlined provisioning of Elasticsearch as a service.

HERE Technologies

May 2019 – Jul 2019

SWE Intern

Mumbai, MH

- Designed a bulk-upload POI spell-checking service **saving 100+ hours** for the map creation team.
- Ideated a proof of concept for stitching Here vehicle street images, enhancing perception for autonomous vehicles.

PROJECTS

Fault-Tolerant Distributed Key-Value Store with Paxos | Go, RPC, Paxos

Jan 2025 – Mar 2025

- Designed a log-replicated distributed key-value store using **Paxos** to ensure **100% consistency** across replicas.
- Reduced duplicate client execution by **95%** via idempotence checks using unique client operation identifiers.
- Achieved **99.9% uptime** and fault-tolerance by recovering and synchronizing lagging replicas using consensus replays.

Hybrid GA + RL XSS Fuzzer for DOM-Based Attacks | Python, Selenium

Jan 2025 – May 2025

- Proposed a hybrid XSS fuzzer using Genetic Algorithms and RL to expose DOM-based vulnerabilities.
- Designed a risk-aware fitness function rewarding stealthy DOM changes, beyond simple alert triggers.

Systolic Array Optimization for Neural Networks | ScaleSimV2

Oct 2024 – Dec 2024

- Tuned systolic arrays with ScaleSimV2 to optimize performance, utilization, and area-delay for neural networks.
- Identified 64×64 array with Output Stationary dataflow as optimal across AlexNet, YOLO Tiny, and MobileNet variants.

CPU Architecture Sensitivity Analysis using GEM5 | C++, GEM5

Sep 2024 – Oct 2024

- Simulated 30+ GEM5 CPU-memory configs using X86TimingSimple & X86Minor with varied clocks and RAM types.
- Benchmarked Stanford Quicksort (273KB working set) with 256KB L2 cache, 1.7B warmup ticks and 4.45T total ticks.
- Analyzed cache miss latency, access latency & power across CPU/memory configs to inform architecture sensitivity.

PUBLICATIONS

K.R. Rathod, S. Bansal, Y.H.C. Kandpal and M. Bisi *Learning Automata-based Feature Selection for Duplex Output Software Effort Estimation Model*, 12th ICCNT, Kharagpur, India, 2021. [Link]